CRPA ANTI-JAMMING BOARD — FIELD TEST REPORT

Date of test	**28-Oct-2026 (Wed)**	◄ Back to Main page
Report / Test ID	(set on Main page)	

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1. Test Information

Test Details			
Test location (Lat / Long)	28.5118865/77.4201639	**Weather / conditions**	Clear Sky
CRPA board S/N		**GNSS receiver model**	
CRPA type ▼	Digital		
Digital CRPA – configuration			
CRPA SDR board type	AntSDR		
DOA method			
Element spacing (mm / λ)	9.52 cm		
Number of elements	2		
Firmware version	Vitis IDE 2023.2		
Bandwidth (MHz)	56 MHz		
Sampling rate (MSPS)	61.44 MS/s		

2. Jammer Parameters

Jammer Configuration			
Jammer type ▼	CW		
Center frequency (MHz)	1582.5	**Tx power (dBm)**	Up-to 40 dB
Spatial angle – Azimuth (°)		**Spatial angle – Elevation (°)**	
Tx antenna gain (dBi)		**Distance to DUT (m)**	6.5m
Est. J/S at DUT (dB)			
CW – no additional parameters (no bandwidth)			

Azimuth measured clockwise from the array reference axis; elevation measured from the horizon.

3. Baseline – Jammer Disabled

Satellite counts and C/N_0 recorded 1 minute after TTFF.

3.1 Pass-Through (Anti-Jamming Algorithm Bypassed)

3.2.1 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	39 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)	7	6	36.83	39
GLONASS (L1OF)	4	4	36.75	39
BeiDou (B1I)	0	0	0	0
Galileo (E1)	5	5	35.2	37
Total / overall (auto)	**16**	**15**	**36.26**	**39**

3.1.2 Hot Start				
TTFF lock time (s) – Hot Start	2 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)	6	6	37.33	39
GLONASS (L1OF)	4	4	38	39
BeiDou (B1I)	0	0	0	0
Galileo (E1)	5	5	35	37
Total / overall (auto)	**15**	**15**	**27.6**	**39**

3.1.3 Cold Start				
TTFF lock time (s) – Cold Start	53	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)	6	6	37.33	39
GLONASS (L1OF)	4	4	37.5	39
BeiDou (B1I)	0	0	0	0
Galileo (E1)	5	5	35.6	38
Total / overall (auto)	**15**	**15**	**27.6**	**39**

3.2 Anti-Jamming Algorithm Enabled

3.2.1 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	74 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)	9	9	36.88	42
GLONASS (L1OF)	5	5	38	41
BeiDou (B1I)	8	8	38.87	43
Galileo (E1)	6	6	35.17	39
Total / overall (auto)	**28**	**28**	**37.2**	**43**

3.2.2 Hot Start				
TTFF lock time (s) – Hot Start	Within 1 Second	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)	8	8	37.38	41
GLONASS (L1OF)	5	5	39	43

CRPA ANTI-JAMMING BOARD — FIELD TEST REPORT				
Date of test	**28-Oct-2026 (Wed)**		◄ Back to Main page	
Report / Test ID	(set on Main page)			

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BeiDou (B1I)	7	7	39.86	44
Galileo (E1)	6	6	35.83	39
Total / overall (auto)	**26**	**26**	**38**	**44**

3.2.3 Cold Start				
TTFF lock time (s) – Cold Start	36 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	8	8	37.37	39
GLONASS (L1OF)	5	5	38.4	43
BeiDou (B1I)	8	8	39	44
Galileo (E1)	6	6	36.33	39
Total / overall (auto)	**27**	**27**	**37.8**	**44**

4. Jammer Enabled – Test 1 – CRPA Nulling Active

Jammer Tx power – Test 1 (dBm) (from Section 2)	15	**Est. J/S at DUT (dB)**	

4.1 Interference Suppression		
Method	**Suppression (dB)**	**Remarks**
RMS (power before vs after nulling)		
Peak (Pmax − Pmin)		

4.2 Electronic Phase Angle / Weights	
Parameter	**Value**
Phase (°)	
Weight magnitude	
Complex weight (I + jQ)	

4.3 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	55 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	36	38
GLONASS (L1OF)	4	3	33.33	35
BeiDou (B1I)	1	1	33	33
Galileo (E1)	2	2	35.5	36
Total / overall (auto)	**13**	**12**	**34.5**	**38**

4.4 Hot Start				
TTFF lock time (s) – Hot Start	Within 1 Second	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	36	38
GLONASS (L1OF)	4	4	32.25	35
BeiDou (B1I)	1	1	33	33
Galileo (E1)	3	3	34.33	36
Total / overall (auto)	**14**	**14**	**33.9**	**38**

4.5 Cold Start				
TTFF lock time (s) – Normal Start	55 Seconds	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	36	38
GLONASS (L1OF)	4	3	33.33	35
BeiDou (B1I)	1	1	33	33
Galileo (E1)	2	2	35.5	36
Total / overall (auto)	**13**	**12**	**34.5**	**38**

4B. Jammer Enabled – Test 2 – CRPA Nulling Active

Jammer Tx power – Test 2 (dBm)	20	**Est. J/S at DUT (dB)**	

4B.1 Interference Suppression		
Method	**Suppression (dB)**	**Remarks**
RMS (power before vs after nulling)		
Peak (Pmax − Pmin)		

4B.2 Electronic Phase Angle / Weights	
Parameter	**Value**
Phase (°)	
Weight magnitude	
Complex weight (I + jQ)	

4B.3 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	178	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	32.17	36
GLONASS (L1OF)	1	1	29	29
BeiDou (B1I)	0	0	0	0
Galileo (E1)	1	1	32	32
Total / overall (auto)	**8**	**8**	**23.3**	**36**

4B.4 Hot Start

CRPA ANTI-JAMMING BOARD — FIELD TEST REPORT			
Date of test	**28-Oct-2026 (Wed)**	◄ Back to Main page	
Report / Test ID	(set on Main page)		

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TTFF lock time (s) – Hot Start	Within 1 Second	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	33.5	36
GLONASS (L1OF)	1	1	30	30
BeiDou (B1I)	0	0	0	0
Galileo (E1)	3	3	32	32
Total / overall (auto)	**10**	**10**	**23.9**	**36**

4B.5 Cold Start				
TTFF lock time (s) – Cold Start	152	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	6	6	33.5	37
GLONASS (L1OF)	2	2	29	30
BeiDou (B1I)	0	0	0	0
Galileo (E1)	3	3	32	32
Total / overall (auto)	**11**	**11**	**23.6**	**37**

4C. Jammer Enabled – Test 3 – CRPA Nulling Active

Jammer Tx power – Test 3 (dBm)	30	**Est. J/S at DUT (dB)**	

4C.1 Interference Suppression		
Method	**Suppression (dB)**	**Remarks**
RMS (power before vs after nulling)		
Peak (Pmax − Pmin)		

4C.2 Electronic Phase Angle / Weights	
Parameter	**Value**
Phase (°)	
Weight magnitude	
Complex weight (I + jQ)	

4C.3 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	No Staellite Lock In	**Position error (m)**	No Staellite Lock In	
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	3	0	29.75	31
GLONASS (L1OF)				
BeiDou (B1I)				
Galileo (E1)				
Total / overall (auto)	**3**	**0**	**29.8**	**31**

4C.4 Hot Start				
TTFF lock time (s) – Hot Start		**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)				
GLONASS (L1OF)				
BeiDou (B1I)				
Galileo (E1)				
Total / overall (auto)				

4C.5 Cold Start				
TTFF lock time (s) – Cold Start		**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)				
GLONASS (L1OF)				
BeiDou (B1I)				
Galileo (E1)				
Total / overall (auto)				

4D. Jammer Enabled – Test 4 – CRPA Nulling Active

Jammer Tx power – Test 4 (dBm)	25	**Est. J/S at DUT (dB)**	

4D.1 Interference Suppression		
Method	**Suppression (dB)**	**Remarks**
RMS (power before vs after nulling)		
Peak (Pmax − Pmin)		

4D.2 Electronic Phase Angle / Weights	
Parameter	**Value**
Phase (°)	
Weight magnitude	
Complex weight (I + jQ)	

4D.3 Normal Start (as powered)				
TTFF lock time (s) – Normal Start	189	**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N$_0$ (dB-Hz)**	**Max C/N$_0$ (dB-Hz)**
GPS (L1 C/A)	4	4	29.75	31
GLONASS (L1OF)	0	0	0	0
BeiDou (B1I)	0	0	0	0
Galileo (E1)	0	0	0	0

CRPA ANTI-JAMMING BOARD — FIELD TEST REPORT				
Date of test	**28-Oct-2026 (Wed)**		◄ Back to Main page	
Report / Test ID	(set on Main page)			

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Total / overall (auto)	**4**	**4**	**7.4**	**31**

4D.4 Hot Start				
TTFF lock time (s) – Hot Start		**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)				
GLONASS (L1OF)				
BeiDou (B1I)				
Galileo (E1)				
Total / overall (auto)				

4D.5 Cold Start				
TTFF lock time (s) – Cold Start		**Position error (m)**		
TTFF lock time (s) + 1 minute		**Fix type / HDOP**		
Constellation	**Satellites tracked**	**Satellites used in fix**	**Avg C/N_0 (dB-Hz)**	**Max C/N_0 (dB-Hz)**
GPS (L1 C/A)				
GLONASS (L1OF)				
BeiDou (B1I)				
Galileo (E1)				
Total / overall (auto)				

5. Performance Summary (auto-calculated)

Jammer Disabled vs Enabled

Parameter	Jammer OFF – Pass-through	Jammer OFF – AJ enabled	Jammer ON – AJ, Test 1 (15 dBm)	Δ (Test 1 − OFF AJ)	Jammer ON – AJ, Test 2 (20 dBm)	Δ (Test 2 − OFF AJ)	Jammer ON – AJ, Test 3 (30 dBm)	Δ (Test 3 − OFF AJ)	Jammer ON – AJ, Test 4 (25 dBm)	Δ (Test 4 − OFF AJ)
Normal start TTFF (s)	39 Seconds	74 Seconds	55 Seconds		178		No Staellite Lock In		189	
Hot start TTFF (s)	2 Seconds	Within 1 Second	Within 1 Second		Within 1 Second					
Cold start TTFF (s)	53	36 Seconds	55 Seconds		152					
Normal start position error (m)							No Staellite Lock In			
Hot start position error (m)										
Cold start position error (m)										
Total satellites tracked (normal start)	16	28	13	**-15**	8	**-20**	3	**-25**	4	**-24**
Total satellites used in fix (normal start)	15	28	12	**-16**	8	**-20**	0	**-28**	4	**-24**
Avg C/N_0, all constellations (dB-Hz, normal start)	36.26	37.2	34.5	**-2.7**	23.3	**-13.9**	29.8	**-7.4**	7.4	**-29.8**
Interference suppression (dB, RMS)	—	—		—		—		—		—

6. Observations & Remarks

7. Sign-off

Role	Name	Signature	Date
Tested by	Sandeep		28/10/2026
Author			
Approved by			